

Grade 3 Ontario Curriculum Spatial Sense

Map It, Move It — Grade 3 Geometry

Unit Overview: Map It, Move It — Grade 3 Geometry

Grade: Grade 3 | **Strand:** Spatial Sense

Target Expectations: E1.1, E1.2, E1.3, E1.4

Learning Goal: Children will sort 3D solids by attributes, compose and decompose structures, identify congruent parts, and give and follow movement directions with turns.

Lesson	Lesson Title	Learning Goal
Day 1	Sort the 3D Solids	I can sort 5 named solids by faces, edges, and vertices.
Day 2	Compose and Decompose	I can compose and decompose a structure into its parts.
Day 3	Congruent Faces	I can identify congruent faces, lengths, and angles.
Day 4	Map It, Move It	I can give and follow movement directions with turns.
Day 5	The Big Map Capstone	I can sort, build, match, and follow movement directions.

Lesson 1: Sort the 3D Solids

Goal: 5 named solids: cubes, prisms, pyramids, cylinders, cones. Sort by FACES, EDGES, VERTICES.

Expectation: E1.1

Learning Goal: "I can sort 5 named solids by faces, edges, and vertices."

Materials: M1_solids_set, M2_attribute_chart, M7_geometry_chart, M10_unit_poster

1. Minds On (12 mins): Trekker Tig's 5 Solids

Trekker Tig: Geometers, today we meet 5 NAMED solids — cubes, prisms, pyramids, cylinders, cones. Count their FACES, EDGES, and VERTICES! Tig opens: 'Sort by ATTRIBUTES, not by LOOKS!'

Bring children to the carpet. Trekker Tig the Tiger Cub puppet with backpack. 'Geometer friends — in Grade 3 we name 5 solids. Each one has a different number of FACES, EDGES, and VERTICES.'

2. Action (25 mins): Sort 15 Solid Cards

- Hand out Solid Card Sets.
- Resort by FACES (1 face, 5 faces, 6 faces).
- Stretch: sort by VERTICES (0, 1, 5, 8).
- Help with cone (curved counts as 1 face?).
- Circulate to record diagnostic.
- GEOMETER'S CHOICE: Pairs INVENT their own sorting rule (by EDGES, curved-vs-flat, tall-vs-short).

3. Consolidation (13 mins): Trekker Tig's Sort Brag

Discuss:

- Name all 5 solids.
- How many faces on a CUBE?
- Where's the VERTEX of a CONE?
- How are CYLINDER and CONE different?
- Tig's preview: tomorrow Coco helps us BUILD with these solids (Lesson 2 — E1.1).

Exit: Tig + Coco refrain — 'Sort by ATTRIBUTES, not by LOOKS — FACES, EDGES, VERTICES!' (3x).

Lesson 2: Compose and Decompose

Goal: Bigger structures are MADE OF smaller 2D shapes and 3D objects.

Expectation: E1.2

Learning Goal: "I can compose and decompose a structure into its parts."

Materials: M1_solids_set, M3_structure_set, M7_geometry_chart, M10_unit_poster

1. Minds On (12 mins): Trekker Tig's Tower Build

Trekker Tig: Watch me COMPOSE a tower from 3 parts. Then DECOMPOSE — what's inside? Coco opens: 'BUILD it BIG, BREAK it small — both ways are MATH!'

Bring children to the carpet. Trekker Tig with backpack and 3 paper solid cards. 'Geometers — today we COMPOSE (build) and DECOMPOSE (break) structures.'

2. Action (25 mins): Compose 4 + Decompose 4

- Hand out Solid Cards + Structure Cards.
- Pairs decompose 4 structure cards: list the parts.
- Stretch: identify 2D shapes inside one 3D part.
- Help with naming — point at chart.
- Circulate to record formative.
- STRATEGY COMPARE 'Two ways to build it': Pairs compose the SAME target structure two different ways (e.g., 4-cube tower vs 4-cube L-shape both = 4 cubes).

3. Consolidation (13 mins): Trekker Tig's Build Brag

Discuss:

- What's COMPOSE?
- What's DECOMPOSE?
- Name 2 parts in a 'house' structure.
- What 2D shape is in a CYLINDER?

Exit: Coco's compose refrain — 'Coco says: BUILD it BIG, BREAK it small — both ways are MATH!' (3x).

Lesson 3: Congruent Faces

Goal: CONGRUENT means SAME SIZE and SAME SHAPE. We match faces, lengths, and angles.

Expectation: E1.3

Learning Goal: "I can identify congruent faces, lengths, and angles."

Materials: M4_congruent_set, M7_geometry_chart, M10_unit_poster

1. Minds On (12 mins): Compass Coco's Match Game

Compass Coco: I have 2 face cards. Are they CONGRUENT? Same size AND same shape? Tig opens: 'SAME SHAPE, SAME SIZE — that's CONGRUENT!'

Bring children to the carpet. Compass Coco the Cat puppet with compass. 'Geometers — today we learn CONGRUENT.'

2. Action (25 mins): Match 10 Congruent Pairs

- Hand out Match Card Sets.
- Identify 2 NON-congruent pairs (close but not match).
- Stretch: explain 1 reason a pair is NOT congruent.
- Help by demonstrating overlap.
- Circulate to record formative.

3. Consolidation (13 mins): Compass Coco's Match Brag

Discuss:

- What's CONGRUENT?
- Are 2 cubes always congruent?
- How do you TEST?
- Find 2 congruent faces in our classroom.

Exit: Tig's congruent refrain — 'SAME SHAPE, SAME SIZE — that's CONGRUENT!' (3x).

Lesson 4: Map It, Move It

Goal: MULTISTEP MOVEMENT uses STEPS plus QUARTER (90°) or HALF (180°) turns.

Expectation: E1.4

Learning Goal: "I can give and follow movement directions with turns."

Materials: M5_map_grid, M8_turn_chart, M9_robot_cards, M7_geometry_chart

1. Minds On (12 mins): Compass Coco's Map Move

Compass Coco: I'll move a marker on the map. 3 steps + quarter turn + 2 steps = treasure! Tig opens: 'Quarter is 90, half is 180 — that's ROTATION!'

Bring children to the carpet. Compass Coco the Cat puppet with compass. 'Geometers — today we MAP IT and MOVE IT.'

2. Action (25 mins): Follow 4 + Give 2

- Hand out Map Grids + 4 direction cards.
- Pairs CREATE 2 multistep directions and trade.
- Stretch: write a 5-step direction with 2 turns.
- Help with quarter vs half turn — body demo.
- Circulate to record formative.
- STRATEGY COMPARE: Pose ONE start/treasure pair on anchor map.

3. Consolidation (13 mins): Compass Coco's Move Brag

Discuss:

- What's a QUARTER TURN?
- What's a HALF TURN?
- How do steps and turns combine?
- Give a 3-step direction to your partner.

Exit: Tig + Coco rotation refrain — 'Quarter is 90, half is 180 — that's ROTATION!' (3x).

Lesson 5: The Big Map Capstone

Goal: Geometers SORT solids, BUILD structures, MATCH congruent parts, and FOLLOW maps.

Expectation: E1.1, E1.2, E1.3, E1.4

Learning Goal: "I can sort, build, match, and follow movement directions."

Materials: M1_solids_set, M2_attribute_chart, M3_structure_set, M4_congruent_set

1. Minds On (12 mins): Trekker Tig's Big Map

Trekker Tig: Today's BIG case — sort, build, match, follow. The 4 detective tools of geometry! Tig + Coco open: 'SORT, COMPOSE, CONGRUENT, MAP — that's GEOMETRY!'

Bring children to the carpet. Trekker Tig and Compass Coco puppets in capes. 'Geometer friends — today is our CAPSTONE. We use ALL we know: 5 solids, compose-decompose, congruent, and movement.'

2. Action (25 mins): The Big Map

- Hand out templates.
- Stations 1-4: sort, build, match, follow.
- Stretch: write a 'geometer's brag': 1 sentence per station.
- Help with quarter/half turn at Station 4.
- Circulate to record summative evidence.
- STRATEGY COMPARE at Station 2 (BUILD) or Station 3 (MATCH): Two geometers use different congruence-checking approaches — overlay vs measure.

3. Consolidation (13 mins): Tig and Coco's Big Map Brag

Discuss:

- What are the 5 solids?
- What's COMPOSE?
- What's CONGRUENT?
- What's a QUARTER TURN?

Exit: Capstone refrain — Tig + Coco: 'SORT, COMPOSE, CONGRUENT, MAP — that's GEOMETRY!' Class chants 3x.

Name: _____

Date: _____

Trekker Tig's 3D Sort

Learning Goal: I can sort 5 named solids by faces, edges, and vertices.

Instructions for the Student:

1. Look at 5 solid icons.
2. For 3 solids, count FACES, EDGES, and VERTICES.
3. Sort 8 mixed solid icons into 5 named piles.

Part 2: Count attributes.

For 3 solids, count FACES, EDGES, and VERTICES.

Date: _____

Tig's Build and Break

Learning Goal: I can compose and decompose a structure into its parts.

Instructions for the Student:

1. For 4 structure pictures, list the 3D parts you see.
2. Build (or draw) 2 structures using 3 parts each.
3. What 2D shapes are **INSIDE** one named 3D solid?

Part 2: Compose 2 structures.

Build (or draw) 2 structures using 3 parts each. List the parts.

[illegible]

(3 parts on the printable worksheet)

Name: _____

Date: _____

Coco's Congruent Match

Learning Goal: I can identify congruent faces, lengths, and angles.

Instructions for the Student:

1. Draw lines from each shape on the left to its CONGRUENT match on the right.
2. For 4 shape pairs, circle YES if congruent, NO if not.
3. Pick ONE pair you said was NOT congruent.

Part 1: Match 5 pairs.

Draw lines from each shape on the left to its CONGRUENT match on the right.

Name: _____

Date: _____

Coco's Map Moves

Learning Goal: I can give and follow movement directions with turns.

Instructions for the Student:

1. Start at A.
2. For 4 turn arrows, write QUARTER or HALF.
3. Write a 4-step direction with at least 1 turn.

Coordinate Grid (8 × 6)

	0	1	2	3	4	5	6	7
5								
4								
3								
2								
1								
0								

Name: _____

Date: _____

Tig and Coco's Big Map

Learning Goal: I can sort, build, match, and follow movement directions.

Instructions for the Student:

1. Name 4 solids and write number of FACES for each.
2. Decompose 1 structure into 3 parts.
3. Start at A.

Part 2: Decompose + match.

Decompose 1 structure into 3 parts. Find 1 congruent pair of faces.

[illegible]

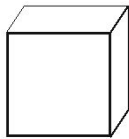
(3 parts on the printable worksheet)

3D Solids Card Set

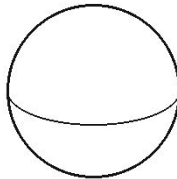
(M1_solids_set)

15 paper 3D solid cards (3 each of cube, prism, pyramid, cylinder, cone).

3D Solids



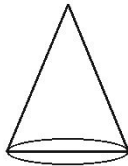
cube



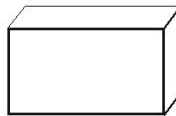
sphere



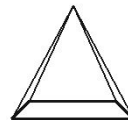
cylinder



cone



rect prism



pyramid

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 15 cards per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 1, 2, 5

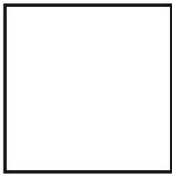
Prep: ~25 min

Faces-Edges-Vertices Chart

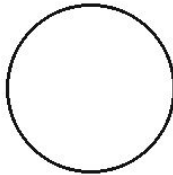
(M2_attribute_chart)

Anchor chart with 5 solids and counts for faces, edges, vertices.

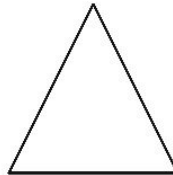
2D Shapes



square



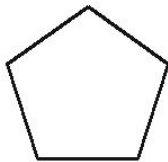
circle



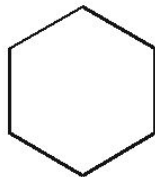
triangle



rectangle



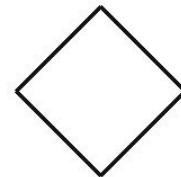
pentagon



hexagon



trapezoid



rhombus

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape to front wall before Lesson 1.

USE IN CLASS

Quantity: 1 chart

Lessons: 1, 5

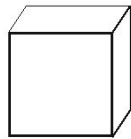
Prep: ~10 min

Structure Build Cards

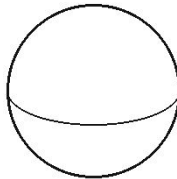
(M3_structure_set)

12 structure cards showing combined 3D objects.

3D Solids



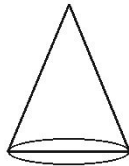
cube



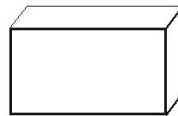
sphere



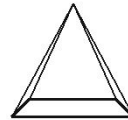
cylinder



cone



rect prism



pyramid

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 12 cards per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 2, 5

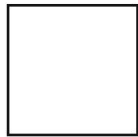
Prep: ~20 min

Congruent Face Match Cards

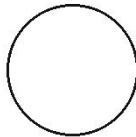
(M4_congruent_set)

20 face cards in 10 congruent pairs.

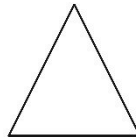
2D Shapes



square



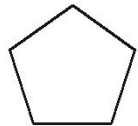
circle



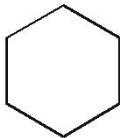
triangle



rectangle



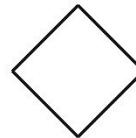
pentagon



hexagon



trapezoid



rhombus

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 20 cards per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 3, 5

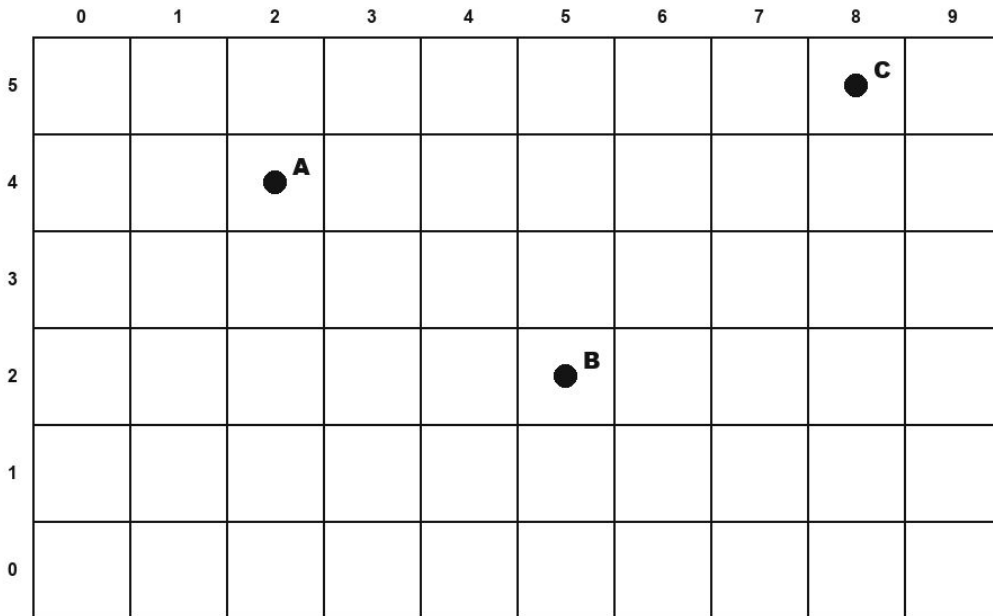
Prep: ~25 min

Movement Map Grid

(M5_map_grid)

Letter-size 6x6 grid map with start and end markers.

Coordinate Grid (10 × 6)



PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 12 maps.

2. Laminate.

3. Distribute one per pair.

USE IN CLASS

Quantity: 10-12 maps

Lessons: 4, 5

Prep: ~15 min

Big Map Template

(M6_capstone_template)

Letter-size capstone template with 4 sections.

Big Map Template	
1	Sort _____
2	Build _____
3	Match _____
4	Follow _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

TEACHER PREP

1. Print 25 templates.

2. Stack near teacher table.

3. No lamination.

USE IN CLASS

Quantity: 25 templates

Lessons: 5

Prep: ~5 min

G3 Geometry Vocabulary Chart

(M7_geometry_chart)

Anchor chart with G3 geometry vocabulary.

G3 Geometry Vocabulary Chart	
Face	Pyramid
• Edge • Vertex • Prism	• Compose • Decompose • Congruent

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape to front wall before Lesson 1.

USE IN CLASS

Quantity: 1 chart

Lessons: 1, 2, 3, 4, 5

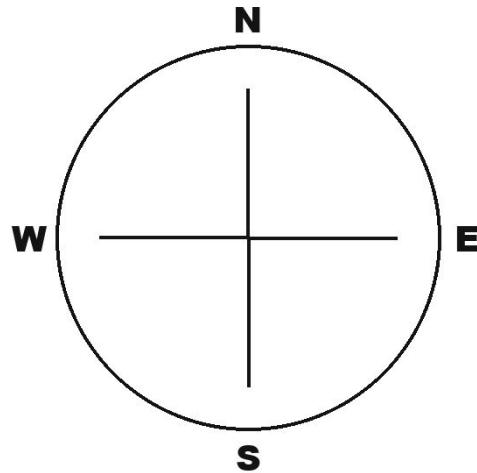
Prep: ~10 min

Quarter and Half Turn Chart

(M8_turn_chart)

Anchor chart of quarter and half turns.

Direction Arrows



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape next to vocab chart.

USE IN CLASS

Quantity: 1 chart

Lessons: 4, 5

Prep: ~10 min

Robot Movement Cards

(M9_robot_cards)

24 movement cards with arrows and step counts.

Coordinate Grid (8 × 6)

	0	1	2	3	4	5	6	7
5								
4								
3								
2								
1								
0								

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 24 cards per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

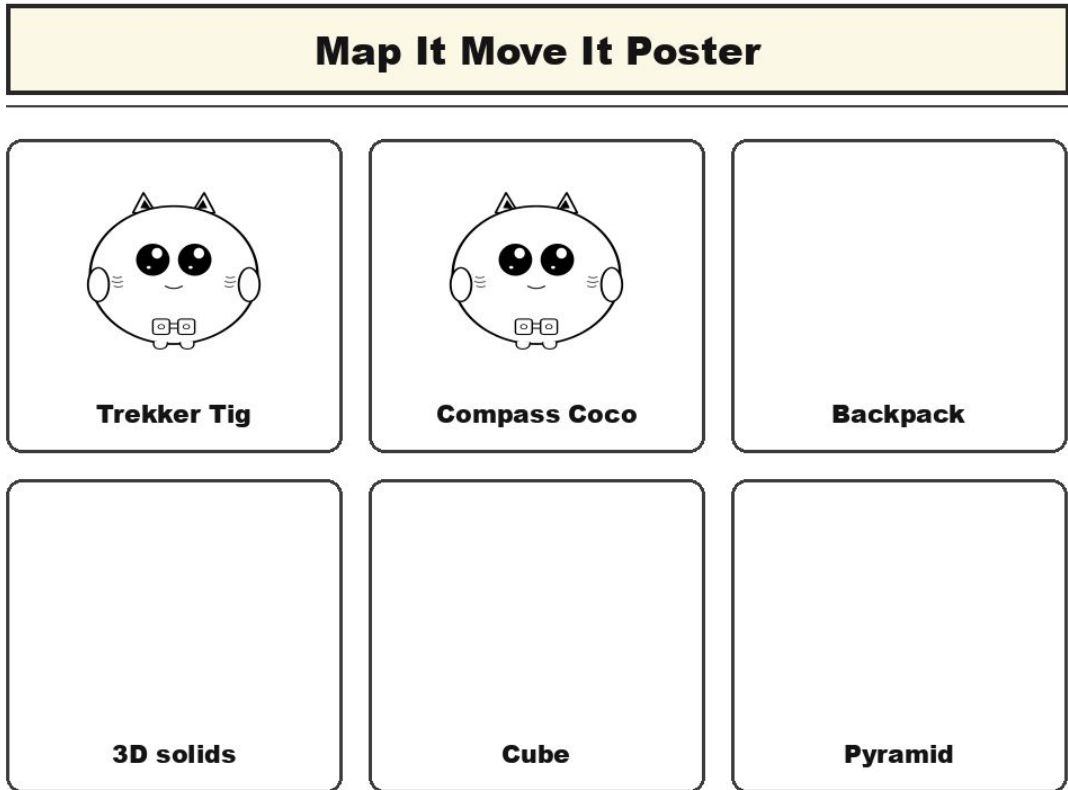
Lessons: 4, 5

Prep: ~25 min

Map It Move It Poster

(M10_unit_poster)

Unit anchor poster of Trekker Tig and Compass Coco.



PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy on letter or 11x17.

2. Laminate.

3. Tape to front wall on Day 1.

USE IN CLASS

Quantity: 1 poster

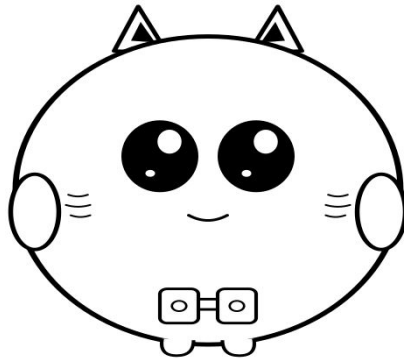
Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

Trekker Tig Puppet

(char_trekker_puppet)

Paper puppet of Trekker Tig the Tiger Cub for teacher use in all 5 lessons.



Trekker Tig the Tiger Cub

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette and tab.

3. Laminate.

4. Glue popsicle stick.

USE IN CLASS

Quantity: 1 puppet

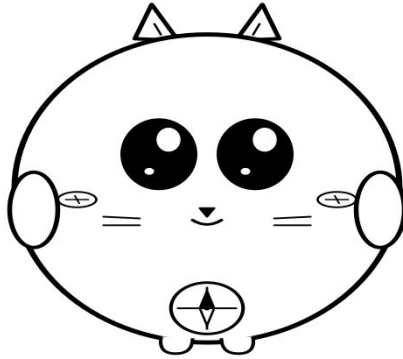
Lessons: 1, 2, 3, 4, 5

Prep: ~12 min

Compass Coco Puppet

(char_compass_puppet)

Paper puppet of Compass Coco the Cat for teacher use in Lessons 3-5.



Compass Coco the Cat

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette and tab.

3. Laminate.

4. Glue popsicle stick.

USE IN CLASS

Quantity: 1 puppet

Lessons: 3, 4, 5

Prep: ~12 min

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
E1.1 sort, construct, and identify cubes, prisms, pyramids, cylinders, and cones by comparing their faces, edges, vertices, and angles	Names 1-2 solids. Wrong attribute counts.	Names 3-4 solids with prompts. Some counts correct.	Independently names all 5 solids and counts faces, edges, vertices.	Sorts by 2 attributes and explains relationships.
E1.2 compose and decompose various structures, and identify the two-dimensional shapes and three-dimensional objects that these structures contain	Cannot identify parts. Builds randomly.	Identifies 1-2 parts in a structure with prompts.	Independently composes structures and decomposes into named parts.	Identifies 2D shapes inside 3D parts.
E1.3 identify congruent lengths, angles, and faces of three-dimensional objects by mentally and physically matching them, and determine if the objects are congruent	Says congruent for 'looks similar'. No size check.	Matches some pairs with prompts.	Independently identifies congruent pairs by checking BOTH size and shape.	Articulates congruent vs similar distinction.
E1.4	Cannot follow	Follows 1-step	Independently	Designs original 4+

Junior Geometer Certificate (G3)

This certificate is proudly presented to:

For successfully completing the 5-Day G3 Map It Move It unit and demonstrating the skills of a G3 Junior Geometer. Trekker Tig the Tiger Cub congratulates the student.

By completing this unit, this student has demonstrated:

- Sorting 5 named 3D solids
- Composing and decomposing structures
- Identifying congruent faces and lengths
- Following multistep movement directions

Marketplace Listing

A 5-day Grade 3 unit on geometry framed around Trekker Tig and Compass Coco.

Price: \$8.99 CAD

Pages: ~45

Classroom time: 250 min

Teacher prep: ~200 min

Top tags: Grade 3, math, geometry, 3D solids, congruent, compose decompose, movement, quarter turn

What's included:

- Unit blueprint (1 file)
- 5 lesson plans (5 files)
- 5 worksheets (5 files)
- Manipulatives plan (1 file)
- Formative + reflection (1 file)
- Assessment suite (1 file)